# PES UNIVERSITY, BANGALORE

**Department of Computer Science and Engineering**

Software Engineering (UE24CS341A) — Mini Project (Jackfruit Phase-1)

## PROJECT SYNOPSIS / PROPOSAL

## 1. PROJECT TITLE AND IDENTIFICATION

| Field | Details |
|---|---|
| **Project Title** | **PassPulse – Smart Event Discovery, High-Concurrency Ticketing & Anti-Fraud Check-in Platform** |
| **Group Number** | **Team 2** |
| **Project ID** | **PP-2026-T02** |
| **Target Stakeholders** | Event Organizers (Clubs, Meetups, Conferences), Venue Staff & Event Attendees |
| **Technology Stack** | React.js, Node.js (Express), PostgreSQL, Redis, Docker, Jenkins |
| **Complexity Level** | Medium |

## 2. TEAM PROFILE

| Sl. | Student Name | SRN | Project Responsibility |
|---|---|---|---|
| 1 | **Sanchita Sunil** | PES1UG24CS419 | Team Lead \| Event Lifecycle & Discovery Module |
| 2 | **Saatvik Gupta** | PES1UG24CS395 | Core Member \| Authentication & Role Management (RBAC) |
| 3 | **Reet V Porwal** | PES1UG24CS369 | Core Member \| Ticket Booking & Automated Waitlist Engine |
| 4 | **Sambhram M** | PES1UG24CS409 | Core Member \| Dynamic QR Check-in & Attendance Analytics |

*Note: Every team member will actively participate in all software engineering phases (SRS, UML Design, Code Reviews, Unit Testing, and CI/CD).*

## 3. PROBLEM STATEMENT

Event management across academic clubs, technical symposiums, and commercial gatherings currently depends on fragmented tools such as Google Forms, static PDF tickets, and manual spreadsheet roll-calls. This causes several practical problems:

- **Gate Delays and Fake Entries:** Static tickets and screenshots are easily shared among multiple attendees, leading to duplicate admissions and long physical queues at entry gates.
- **Inventory Overbooking:** Without proper database transaction locking, high-traffic registrations cause race conditions where more tickets are issued than the venue capacity.
- **Unused Capacity from No-Shows:** When events are marked sold-out, there is no automatic system to allocate slots from cancellations to waitlisted attendees.
- **Lack of Attendance Visibility:** Organizers do not have real-time data on actual turnouts versus registered attendees during the event.

**PassPulse** provides an integrated web platform to automate event discovery, multi-tier ticket reservation with temporary holds, dynamic QR gate validation, automated waitlists, and live attendance metrics.

## 4. OBJECTIVES

- Provide an easy-to-use event discovery portal with category and date filtering.
- Implement safe, concurrency-controlled ticket booking using atomic database operations to eliminate overselling.
- Generate dynamic, time-sensitive QR code tickets to prevent ticket sharing and fraud.
- Automate the waitlist process so cancelled tickets are automatically offered to the next attendee in line within a fixed claim window.
- Provide a mobile-friendly scanning interface for gate staff to verify tickets and record entries in under 200 ms.
- Follow Agile principles with modular architecture, Git branch workflows, automated unit tests, and continuous integration pipelines.

## 5. INTENDED USERS

1. **Attendees:** Browse events, book tier-based tickets, view digital QR passes, and join waitlists for full events.
2. **Organizers:** Create, schedule, and edit events; manage ticket categories and pricing; scan QR passes at entry; view attendance reports.
3. **Gate Staff:** Use the dedicated in-app scanner to validate attendee passes rapidly at entry gates.
4. **System Admin:** Manage user accounts, approve public event listings, and monitor platform health.

## 6. FUNCTIONAL FEATURES

1. **User Registration & Role-Based Access Control (RBAC):** Multi-role authentication (Attendee, Organizer, Gate Staff, Admin) with secure password hashing and JWT sessions.
2. **Event Catalog & Multi-Parameter Search:** Search and filter events based on category, date, venue, ticket price, and availability.
3. **Event Creation & Lifecycle Management:** Organizer dashboard to create, update, schedule, and cancel events with configurable capacities and venue rules.
4. **Multi-Tier Ticket Management:** Support for multiple ticket types (General, Early Bird, VIP) with independent pricing, release windows, and seat limits.
5. **Ticket Reservation with Temporary Hold:** Concurrency-safe booking workflow that temporarily reserves tickets for 10 minutes during checkout to prevent race conditions.
6. **Dynamic QR Code Pass Generation:** Automated generation of digital QR tickets with cryptographic validation to stop screenshot reuse.
7. **Fast In-App QR Check-in:** Camera-based scanner interface for gate staff that checks pass validity, marks attendance, and rejects duplicate scans.
8. **Automated Waitlist & Slot Reallocation:** FIFO waitlist queue that automatically sends a 15-minute claim notification to the next user when a ticket is cancelled.
9. **Live Attendance Telemetry & Reports:** Real-time charts showing check-in progress, attendance percentages, and downloadable summary reports.
10. **Admin Moderation Portal:** Tools to review new event listings, manage platform users, and review activity logs.

## 7. PLAN OF WORK AND PRODUCT OWNERSHIP

*In accordance with course guidelines, each member has full ownership of one complete functional feature from requirements to validation. Supporting duties (UI, DB, CI/CD, Testing) are distributed as secondary responsibilities.*

| Member | Owned Functional Feature | Secondary Role | Qualitative Target |
|---|---|---|---|
| **Sanchita Sunil** (PES1UG24CS419) | **Event Discovery, Catalog Search & Lifecycle Management** | UI/UX Design & Frontend Integration | Search query response time under 150 ms across event listings. |
| **Saatvik Gupta** (PES1UG24CS395) | **User Authentication, Session & Role-Based Access Control** | API Gateway & Security Validation | Zero high-severity vulnerabilities in SonarQube audit; sub-80 ms token verification. |
| **Reet V Porwal** (PES1UG24CS369) | **Multi-Tier Ticket Booking & Automated Waitlist Engine** | Database Modeling & ACID Transactions | 100% data consistency with 0 overbooking during concurrent bookings. |

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|---|---|---|---|
| **Sambhram M** (PES1UG24CS409) | **Dynamic QR Pass Generation, Gate Check-in & Analytics** | Unit Testing & CI/CD Pipeline (Jenkins) | Gate verification latency under 200 ms; test coverage ≥ 80%. |

## 8. SHORT-TERM AGILE PLAN (SPRINT SCHEDULE)

| Iteration | Duration | Planned Output | Acceptance Evidence |
|---|---|---|---|
| **Sprint 1** | Weeks 1–2 | Requirements baseline, SRS document, Jira user stories, and Git repository setup. | Approved SRS v1.0 + Configured GitHub repository with branch protection rules. |
| **Sprint 2** | Weeks 2–3 | User Authentication, Role Management (RBAC), and basic profile endpoints. | Working Auth API + Unit test cases + Passing SonarQube quality gate. |
| **Sprint 3** | Weeks 3–4 | Event creation wizard, multi-tier pricing setup, and catalog search engine. | Functional event catalog UI + Backend API endpoints + Mock integration tests. |
| **Sprint 4** | Weeks 5–6 | Ticket reservation with temporary hold, booking confirmation, and waitlist logic. | End-to-end booking flow + Concurrency test scripts showing zero overbooking. |
| **Sprint 5** | Weeks 6–7 | Dynamic QR generation, in-app gate scanner module, and real-time attendance dashboard. | Live camera QR verification demo + Gate log verification + ≥ 80% test coverage. |
| **Sprint 6** | Weeks 7–8 | Admin moderation dashboard, report export, Dockerization, and deployment. | Complete Docker-Compose build + Jenkins CI/CD pipeline execution + Final demo. |

## 9. TECHNOLOGY STACK AND TOOLS

- **Frontend:** React.js, Tailwind CSS, Axios, HTML5 QR Scanner.
- **Backend:** Node.js with Express.js REST API.
- **Database & Cache:** PostgreSQL (ACID relational data) and Redis (locks and temporary holds).
- **Version Control:** Git and GitHub (PR reviews and branch protection).
- **Agile Project Management:** Jira (User Stories, Sprint Backlogs, Story Points).
- **CI/CD & Code Quality:** Jenkins / GitHub Actions, SonarQube (Quality Gate compliance).
- **Testing & Containerization:** Jest, Supertest, Postman, Docker & Docker-Compose.

## 10. PROJECT SCOPE

**In-Scope:** User authentication and RBAC; event listing and search; multi-tier ticket management; concurrency-controlled checkout; dynamic QR passes; mobile gate scanning; automated FIFO waitlist reallocation; real-time attendance dashboards and PDF/CSV summary reports.

**Out-of-Scope:** Real banking payment gateway integration (mock payment status will be used); facial recognition entry; live video streaming; indoor venue maps.